The differences between Rails and Phoenix are numerous—they are different frameworks, have different goals, and are built for different programming paradigms! Instead of attempting to list specific differences, this section covers major differences in the goals and development of each framework.

We're covering as much of Phoenix in this chapter as we can, but there's a dedicated book if you want more. Check out *Programming Phoenix 1.4 [TV19]* by Chris McCord (Phoenix's creator), Bruce Tate, and José Valim.

Let's first look at how Phoenix can work with so little code, and then we'll compare Phoenix and Rails. Let's get started!

What Is Phoenix?

Phoenix is a fairly lightweight library, but it has a ton of features. This is due to the core Phoenix library integrating with other libraries to achieve key features. You won't often interface with these other libraries directly, but it's important to know about them in case something goes wrong or you need to change a setting.

Here are the features that Phoenix brings to the table, either directly or through integrations with other libraries:

Serve Web Requests via Controller Modules

Serving web requests is the core feature of Phoenix. Phoenix interfaces with the Plug[1] library to make this possible. Plug consists of a pipeline that lets you modify a web request via composable middleware modules. This is similar to how Rack[2] works in Ruby.

Provide HTML Helpers, Templates, and Components

Phoenix provides the Phoenix.HTML[3] package for HTML-related helper modules. Phoenix also provides a component framework[4] via the `phoenix_live_view` package, so you can easily convert frontend code into components and use them across your application. Phoenix components are becoming more prominent, and they will likely replace `Phoenix.HTML` in the future.

Integrate with a Web Server That Works out of the Box

Phoenix doesn't provide a web server, but it uses Bandit[5] instead. Bandit is a pure-Elixir web server that's built for high concurrency and safety.

1. https://hexdocs.pm/plug/readme.html
2. https://github.com/rack/rack
3. https://github.com/phoenixframework/phoenix_html
4. https://hexdocs.pm/phoenix_live_view/Phoenix.Component.html
5. https://github.com/mtrudel/bandit